

Development Report

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Work Completed

1. Publeader — Authentication Architecture & Implementation

- Conducted technical research and selected Better Auth (v1.6.9) as the authentication framework, evaluated against alternatives for compatibility with Next.js 15 App Router and Expo React Native
 - Designed and implemented a unified authentication backend on Next.js serving both the web dashboard and the mobile application — single source of truth for all user sessions
 - Integrated MongoDB Atlas as the persistent store via the Better Auth MongoDB adapter, with connection singleton to handle Next.js hot-reload safely in development
 - Designed a multi-role user model: `admin`, `driver`, `advertiser`, `partner` — each with an associated status lifecycle (`pending` → `validated` / `rejected`)
 - Built per-role registration API endpoints (`/api/register/driver`, `/api/register/company`, `/api/register/partner`) where the role is assigned server-side — clients cannot self-escalate privileges
 - Implemented email OTP flow (6-digit codes, 10-minute expiry) for both email verification on signup and password reset — using nodemailer with Gmail SMTP and a console fallback for local development
 - Configured the Better Auth admin, organization, and emailOTP plugins with custom additional user fields (role, status, phone, linked entity IDs)
 - Built a `/api/me` hydration endpoint that returns the authenticated user alongside their linked entity document (driver profile, company, or partner) — used by both web and mobile on session restore
 - Implemented dual-client authentication strategy: httpOnly cookies for the web dashboard, `expo-secure-store` via `@better-auth/expo` for the mobile app — ensuring secure token storage on both platforms
 - Wrote the mobile `AuthContext` with full session hydration, role-aware redirects, and pending-status polling (every 15s until account is validated by admin)
 - Built all auth screens on mobile: login, register (driver / company / partner), verify email, forgot password, reset password, pending approval
 - Configured Next.js middleware using `getSessionCookie` to protect all dashboard routes, with matcher rules excluding public pages and API routes
 - Set up environment configuration for Better Auth secret, MongoDB Atlas URI, and Gmail SMTP across `.env.local` and `.env.example`
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2. Publeader — Developer Experience & Tooling

- Built seed scripts (`seed:users`, `seed:admin`) to populate MongoDB with demo accounts for all four roles, each with linked entity documents and pre-validated status — allowing instant testing without manual signup flow

- Created fast-login developer pages on both web (`/dev-login`) and mobile (`/(auth)/dev-login`), gated behind `__DEV__` on mobile — one-tap sign-in for any demo role during development
 - Updated all app identifiers, deep-link schemes, bundle IDs, MongoDB database name, and email domains from legacy `driveads` branding to `publeader` across both repositories
 - Generated and configured a cryptographically secure Better Auth secret (48-byte base64url)
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3. AI Aqd — APK Distribution & Stakeholder Onboarding

- Managed APK distribution to end users: prepared and shared installation links, registered testers in Google Play Console, and provided hands-on installation support
 - Attended two project meetings dedicated to the AI Aqd application — first meeting to receive a full walkthrough of the app's purpose and feature set, second meeting to clarify technical details and align on next steps
 - Acted as the technical point of contact between the development environment and the end users during the onboarding phase
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In Progress

- Publeader: server-side role-based access control middleware on protected API routes
 - Publeader: admin user management UI (approve/reject pending accounts, force-logout, ban/unban)
 - Publeader: team member invite acceptance flow
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